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Кафедра Цифровые технологии управления транспортными процессами»

Отчет по Лабораторной Работе №3

Вариант 13

По дисциплине «Базы данных и экспертные системы»

На тему «SQL (DDL, DCL)»

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I. Цель работы

Приобрести практические навыки проектирования структуры реляционной базы данных с использованием языка определения данных (DDL) и управления правами доступа с помощью языка контроля данных (DCL), а также развить способность обеспечивать целостность, безопасность и актуальность информации в многопользовательской среде.

II. Задание

1 Создать таблицу Residing (Проживающие)

№ п/п	Имя столбца	Описание	Тип данных	Ограничения целостности
1.	id_residing	id проживания	integer	первичный ключ
2.	id_warrant	id ордера проживания	integer	внешний ключ
3.	id_premises	id помещения	integer	
4.	d_start	дата начала	date	пустые значения запрещены
5.	d_end	дата окончания	date	
6.	id_bas_sal_date	дата перерасчета	date	не пусто

```
CREATE TABLE Residing (
id_residing INTEGER PRIMARY KEY,
id_warrant INTEGER,
id_premises INTEGER,
d_start DATE NOT NULL,
d_end DATE,
id_bas_sal_date INTEGER NOT NULL
);
```

```
CREATE TABLE Residing (
id_residing INTEGER PRIMARY KEY,
id_warrant INTEGER,
id_premises INTEGER,
d_start DATE NOT NULL,
d_end DATE,
id_bas_sal_date INTEGER NOT NULL
);
```

Script Output | Query Result | Query Result 1

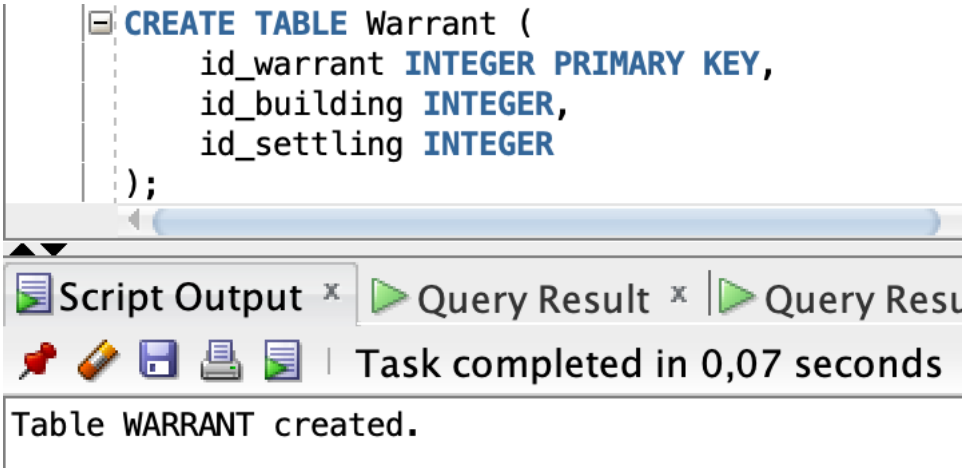
Task completed in 0,07 seconds

```
Table RESIDING created.
```

2 Создать таблицу Warrant (Ордер_проживающего)

№ п/п	Имя столбца	Описание	Тип данных	Ограничения целостности
1.	id_warrant	id ордера проживания	integer	первичный ключ
2.	id_building	id здания	integer	внешний ключ
3.	id_settling	id заселения	integer	

```
CREATE TABLE Warrant (
    id_warrant INTEGER PRIMARY KEY,
    id_building INTEGER,
    id_settling INTEGER
);
```



3 Создать таблицу Building (Здания)

№ п/п	Имя столбца	Описание	Тип данных	Ограничения целостности
1.	id_building	id здания	integer	первичный ключ
2.	b_name	наименование	char(10)	
3.	b_start	дата начала	date	не пусто
4.	b_note	примечание	char(10)	

```
CREATE TABLE Building (
    id_building INTEGER PRIMARY KEY,
    b_name CHAR(100),
    b_start DATE NOT NULL,
    b_note CHAR(100)
);
```

```
CREATE TABLE Building (
    id_building INTEGER PRIMARY KEY,
    b_name CHAR(100),
    b_start DATE NOT NULL,
    b_note CHAR(100)
);
```

Script Output | Query Result | Query

Task completed in 0,07 seco

```
Table BUILDING created.
```

4 Создать таблицу Premises (Помещения)

№ п/п	Имя столбца	Описание	Тип данных	Ограничения целостности
1.	id_premises	id помещения	integer	первичный ключ
2.	p_name	название	varchar(30)	не пусто
3.	p_d_start	дата начала	date	не пусто
4.	id_building	id здания	integer	внешний ключ

```
CREATE TABLE Premises (
    id_premises INTEGER PRIMARY KEY,
    p_name VARCHAR(30) NOT NULL,
    p_d_start DATE NOT NULL,
    id_building INTEGER
);
```

```
CREATE TABLE Premises (
    id_premises INTEGER PRIMARY KEY,
    p_name VARCHAR(30) NOT NULL,
    p_d_start DATE NOT NULL,
    id_building INTEGER
);
```

Script Output | Query Result | Query

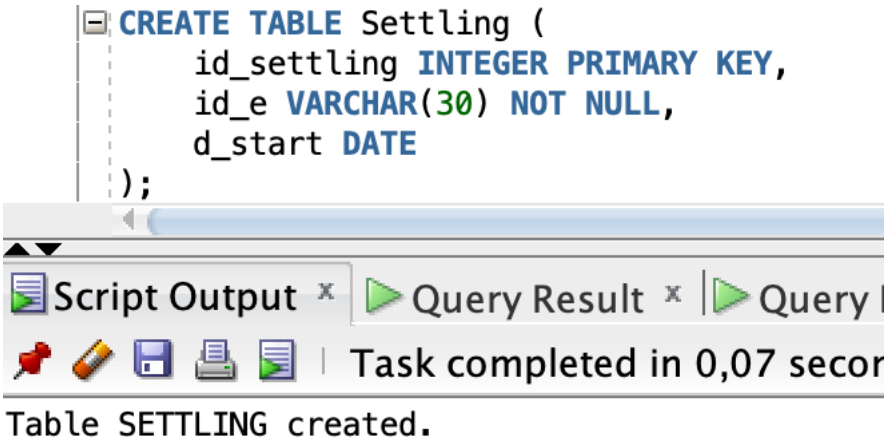
Task completed in 0,07 secor

```
Table PREMISES created.
```

5 Создать таблицу Settling (Заселение)

№ п/п	Имя столбца	Описание	Тип данных	Ограничения целостности
1.	id_settling	id заселения	integer	первичный ключ
2.	id_e	id человека	varchar(30)	не пусто
3.	d_start	дата начала	date	

```
CREATE TABLE Settling (
    id_settling INTEGER PRIMARY KEY,
    id_e VARCHAR(30) NOT NULL,
    d_start DATE
);
```



<u>Теперь когда все таблицы созданы укажем все необходимые внешние ключи:</u>

```
ALTER TABLE  Residing
ADD CONSTRAINT fk_residing_warrant
FOREIGN KEY (id_warrant) REFERENCES Warrant(id_warrant);

ALTER TABLE Residing
ADD CONSTRAINT fk_residing_premises
FOREIGN KEY (id_premises) REFERENCES Premises(id_premises);

ALTER TABLE Warrant
ADD CONSTRAINT fk_warrant_building
FOREIGN KEY (id_building) REFERENCES Building(id_building);

ALTER TABLE Premises
ADD CONSTRAINT fk_premises_building
FOREIGN KEY (id_building) REFERENCES Building(id_building);
```

```
ALTER TABLE  Residing
ADD CONSTRAINT fk_residing_warrant
FOREIGN KEY (id_warrant) REFERENCES Warrant(id_warrant);

ALTER TABLE Residing
ADD CONSTRAINT fk_residing_premises
FOREIGN KEY (id_premises) REFERENCES Premises(id_premises);

ALTER TABLE Warrant
ADD CONSTRAINT fk_warrant_building
FOREIGN KEY (id_building) REFERENCES Building(id_building);

ALTER TABLE Premises
ADD CONSTRAINT fk_premises_building
FOREIGN KEY (id_building) REFERENCES Building(id_building);
```

Script Output | Query Result | Query Result 1

Task completed in 0,07 seconds

```
Table RESIDING altered.


Table RESIDING altered.


Table WARRANT altered.


Table PREMISES altered.
```

6 Указать для таблицы Warrant внешний ключ (id_settling) и задать ограничение –

```
ALTER TABLE Warrant
ADD CONSTRAINT fk_warrant_settling
FOREIGN KEY (id_settling) REFERENCES Settling(id_settling);
ALTER TABLE Warrant
ADD CONSTRAINT check_warrant_gt_settling
CHECK (id_warrant > id_settling);
```

```
ALTER TABLE Warrant
ADD CONSTRAINT fk_warrant_settling
FOREIGN KEY (id_settling) REFERENCES Settling(id_settling);

ALTER TABLE Warrant
ADD CONSTRAINT check_warrant_gt_settling
CHECK (id_warrant > id_settling);
```

Script Output | Query Result | Query Result 1

Task completed in 0,07 seconds

```
Table WARRANT altered.


Table WARRANT altered.
```

7 Добавить в таблицу Building ограничение – наименование здания не должно быть
пустым, по умолчанию «_».

```
ALTER TABLE Building
MODIFY b_name CHAR(10) DEFAULT '_' NOT NULL;
```

```
ALTER TABLE Building
MODIFY b_name CHAR(10) DEFAULT '_' NOT NULL;
```

Script Output | Query Result | Query Result 1

Task completed in 0,07 seconds

```
Table RESIDING altered.


Table RESIDING altered.


Table WARRANT altered.


Table PREMISES altered.
```

8 Вставить в таблицы следующие значения:

1. Building

id_building	b_name	b_note	b_start
11	К-1	Здание учебного корпуса № 1	01.01.2010
17	ЗО-1	Здание общежития № 1	01.01.2010
18	ЗО-2	Здание общежития № 2	01.01.2010

```
INSERT INTO Building (id_building, b_name, b_note, b_start) VALUES
(11, 'К-1', 'Здание учебного корпуса № 1',
TO_DATE('01.01.2010','DD.MM.YYYY'));

INSERT INTO Building (id_building, b_name, b_note, b_start) VALUES
(17, 'ЗО-1', 'Здание общежития № 1', TO_DATE('01.01.2010', 'DD.MM.YYYY'));

INSERT INTO Building (id_building, b_name, b_note, b_start) VALUES
(18, 'ЗО-2', 'Здание общежития № 2', TO_DATE('01.01.2010', 'DD.MM.YYYY'));
```

```
INSERT INTO Building (id_building, b_name, b_note, b_start) VALUES
(18, 'ЗО-2', 'Здание общежития № 2', TO_DATE('01.01.2010', 'DD.MM.YYYY'));
```

Script Output

Task completed in 0,034 seconds

```
1 row inserted.


1 row inserted.
```

```
SELECT * FROM Building;
```

```
SELECT * FROM Building;
```

Script Output | Query Result

SQL | All Rows Fetched: 3 in 0,004 seconds

	ID_BUILDING	B_NAME	B_START	B_NOTE
1	11	К-1	01 ЯНВ 2010	Здание учебного корпуса № 1
2	17	30-1	01 ЯНВ 2010	Здание общежития № 1
3	18	30-2	01 ЯНВ 2010	Здание общежития № 2

2. Settling

id_settling	id_e	d_start
50	1	01.08.2011
51	2	01.09.2011
52	4	15.08.2011

```
INSERT INTO Settling (id_settling, id_e, d_start) VALUES
(50, '1', TO_DATE('01.08.2011', 'DD.MM.YYYY'));

INSERT INTO Settling (id_settling, id_e, d_start) VALUES
(51, '2', TO_DATE('01.09.2011', 'DD.MM.YYYY'));

INSERT INTO Settling (id_settling, id_e, d_start) VALUES
(52, '4', TO_DATE('15.08.2011', 'DD.MM.YYYY'));
```

```
INSERT INTO Settling (id_settling, id_e, d_start) VALUES
(52, '4', TO_DATE('15.08.2011', 'DD.MM.YYYY'));
```

Query Result | Script Output

Task completed in 0,028 seconds

```
1 row inserted.

1 row inserted.

1 row inserted.
```

```
select * from Settling;
```

```
SELECT * FROM Settling;
```

Script Output | Query Result

SQL | All Rows Fetched: 3 in 0

	ID_SETTLING	ID_E	D_START
1	50	1	01 АВГ 2011
2	51	2	01 СЕН 2011
3	52	4	15 АВГ 2011

3. Premises

id_premises	name	p_d_start	p_area	id_building
1	110	22.06.2010	2	17
2	112	22.06.2010	5	18
4	115	22.06.2010	2	18
5	1110	22.06.2010	3	17
7	211	22.05.2010	4	17
8	1111	22.05.2010	50	11

```
ALTER TABLE Premises ADD p_area NUMBER;
```

```
ALTER TABLE Premises ADD p_area NUMBER;
```

Query Result | Script Output

Task completed in 0,098 seconds

```
Table PREMISES altered.
```

```
INSERT INTO Premises (id_premises, p_name, p_d_start, p_area, id_building)
VALUES
(1, '110', TO_DATE('22.06.2010', 'DD.MM.YYYY'), 2, 17);

INSERT INTO Premises (id_premises, p_name, p_d_start, p_area, id_building)
VALUES
(2, '112', TO_DATE('22.06.2010', 'DD.MM.YYYY'), 5, 18);
```

```sql
INSERT INTO Premises (id_premises, p_name, p_d_start, p_area, id_building)
VALUES
(4, '115', TO_DATE('22.06.2010', 'DD.MM.YYYY'), 2, 18);

INSERT INTO Premises (id_premises, p_name, p_d_start, p_area, id_building)
VALUES
(5, '1110', TO_DATE('22.06.2010', 'DD.MM.YYYY'), 3, 17);

INSERT INTO Premises (id_premises, p_name, p_d_start, p_area, id_building)
VALUES
(7, '211', TO_DATE('22.05.2010', 'DD.MM.YYYY'), 4, 17);

INSERT INTO Premises (id_premises, p_name, p_d_start, p_area, id_building)
VALUES
(8, '1111', TO_DATE('22.05.2010', 'DD.MM.YYYY'), 50, 11);
```

```sql
INSERT INTO Premises (id_premises, p_name, p_d_start, p_area, id_building) VALU
(8, '1111', TO_DATE('22.05.2010', 'DD.MM.YYYY'), 50, 11);
```

Script Output

Task completed in 0,055 seconds

```
1 row inserted.


1 row inserted.
```

```sql
SELECT * FROM Premises;
```

```sql
SELECT * FROM Premises;
```

Script Output | Query Result

SQL | All Rows Fetched: 6 in 0,066 seconds

	ID_PREMISES	P_NAME	P_D_START	ID_BUILDING	P_AREA
1	1	110	22 ИЮН 2010	17	2
2	2	112	22 ИЮН 2010	18	5
3	4	115	22 ИЮН 2010	18	2
4	5	1110	22 ИЮН 2010	17	3
5	7	211	22 МАЙ 2010	17	4
6	8	1111	22 МАЙ 2010	11	50

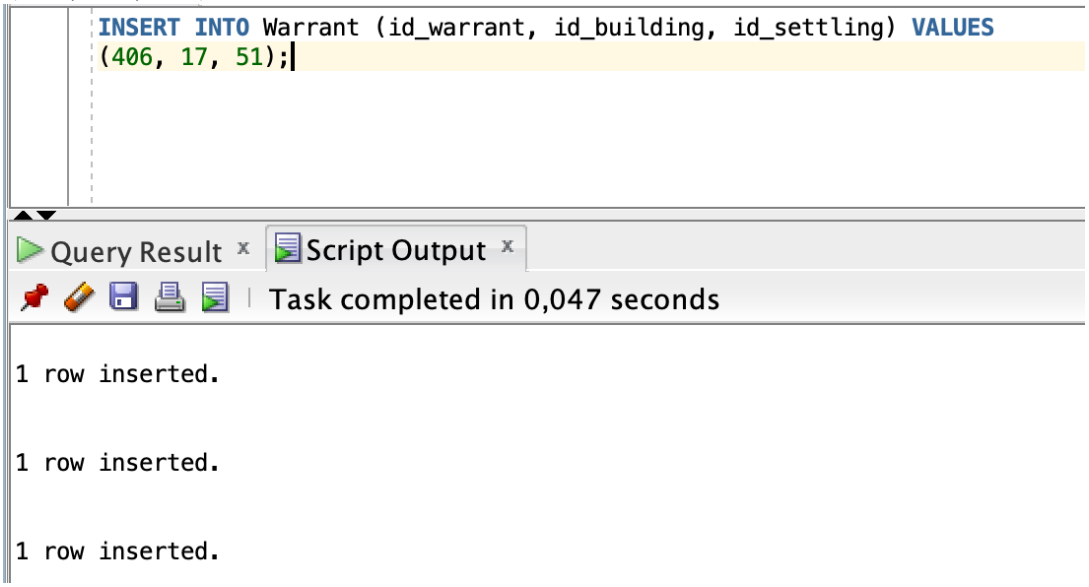
4. Warrant

id_warrant	id_building	id_settling
400	17	50
404	17	52
404	18	52
406	17	51

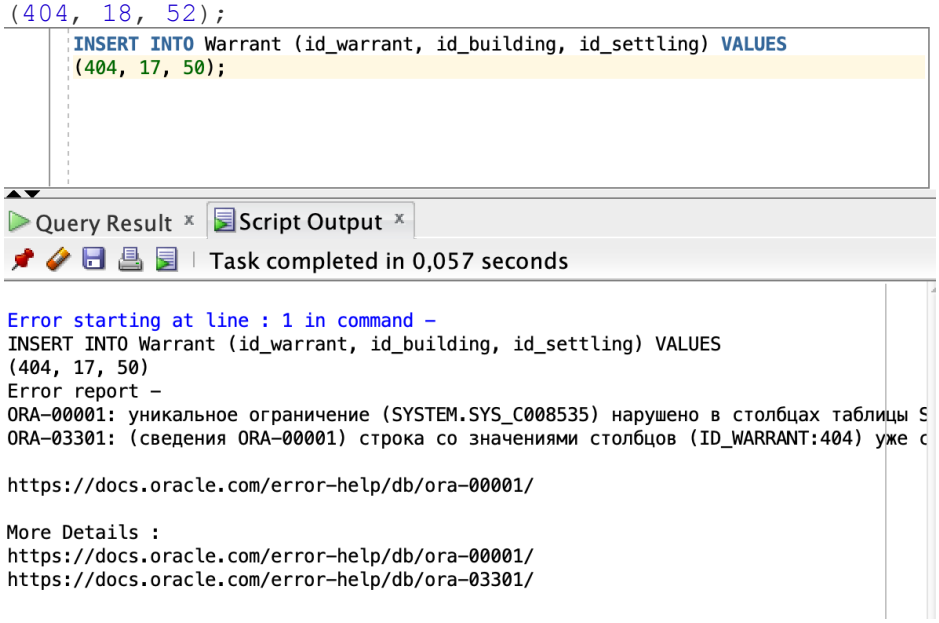
```
INSERT INTO Warrant (id_warrant, id_building, id_settling) VALUES
(400, 17, 50);

INSERT INTO Warrant (id_warrant, id_building, id_settling) VALUES
(404, 17, 52);

INSERT INTO Warrant (id_warrant, id_building, id_settling) VALUES
(406, 17, 51);
```




```
INSERT INTO Warrant (id_warrant, id_building, id_settling) VALUES
(404, 18, 52);
```



select * from Warrant;

```
SELECT * FROM Warrant;
```

Script Output | Query Result

SQL | All Rows Fetched: 3 in 0,009 seconds

	ID_WARRANT	ID_BUILDING	ID_SETTLING
1	400	17	50
2	404	17	52
3	406	17	51

5. Residing

id_residing	id_warrant	id_premises	d_start	d_end	id_bas_sal
301	400	5	02.08.2011	30.09.2011	90
302	404	7	15.08.2011	15.09.2011	91
305	404	2	16.09.2011	30.09.2011	92
306	400	1	01.10.2011		93
307	406	1	01.08.2011		94

```
INSERT INTO Residing (id_residing, id_warrant, id_premises, d_start, d_end,
id_bas_sal_date)
VALUES (301, 400, 5, TO_DATE('02.08.2011', 'DD.MM.YYYY'),
TO_DATE('30.09.2011', 'DD.MM.YYYY'), 90);

INSERT INTO Residing (id_residing, id_warrant, id_premises, d_start, d_end,
id_bas_sal_date)
VALUES (302, 404, 7, TO_DATE('15.08.2011', 'DD.MM.YYYY'),
TO_DATE('15.09.2011', 'DD.MM.YYYY'), 91);

INSERT INTO Residing (id_residing, id_warrant, id_premises, d_start, d_end,
id_bas_sal_date)
VALUES (305, 404, 2, TO_DATE('16.09.2011', 'DD.MM.YYYY'),
TO_DATE('30.09.2011', 'DD.MM.YYYY'), 92);

INSERT INTO Residing (id_residing, id_warrant, id_premises, d_start, d_end,
id_bas_sal_date)
VALUES (306, 400, 1, TO_DATE('01.10.2011', 'DD.MM.YYYY'), NULL, 93);

INSERT INTO Residing (id_residing, id_warrant, id_premises, d_start, d_end,
id_bas_sal_date)
VALUES (307, 406, 1, TO_DATE('01.08.2011', 'DD.MM.YYYY'), NULL, 94);
```

```
INSERT INTO Residing (id_residing, id_warrant, id_premises, d_start, d_end,
VALUES (307, 406, 1, TO_DATE('01.08.2011', 'DD.MM.YYYY'), NULL, 94);
```

Script Output | Query Result

Task completed in 0,027 seconds

```
1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.
```

```
select * from Residing;
```

```
SELECT * FROM Residing;
```

Script Output | Query Result

SQL | All Rows Fetched: 5 in 0,018 seconds

	ID_RESIDING	ID_WARRANT	ID_PREMISES	D_START	D_END	I...
1	301	400	5	02 АВГ 2011	30 СЕН 2011	90
2	302	404	7	15 АВГ 2011	15 СЕН 2011	91
3	305	404	2	16 СЕН 2011	30 СЕН 2011	92
4	306	400	1	01 ОКТ 2011	(null)	93
5	307	406	1	01 АВГ 2011	(null)	94

9 Изменить дату окончания проживания у id_warrant = 406 на «01.06.2012».

```
UPDATE Residing
SET d_end = TO_DATE('01.06.2012', 'DD.MM.YYYY')
WHERE id_warrant = 406;
```

```
UPDATE Residing
SET d_end = TO_DATE('01.06.2012', 'DD.MM.YYYY')
WHERE id_warrant = 406;
```

Script Output

Task completed in 0,045 seconds

```
1 row updated.
```

```
select * from Residing where id_warrant=406;
```

```
SELECT * FROM Residing WHERE id_warrant = 406;
```

Script Output | Query Result

SQL | All Rows Fetched: 1 in 0,009 seconds

	ID_RESIDING	ID_WARRANT	ID_PREMISES	D_START	D_END	ID_BAS_SAL_DATE
1	307	406	1	01 АВГ 2011	01 ИЮН 2012	94

10 Создать пользователей user1, user2, user3, user4.

```
CREATE USER user1 IDENTIFIED BY password1;
CREATE USER user2 IDENTIFIED BY password2;
CREATE USER user3 IDENTIFIED BY password3;
CREATE USER user4 IDENTIFIED BY password4;
```

```
CREATE USER user1 IDENTIFIED BY password1;
CREATE USER user2 IDENTIFIED BY password2;
CREATE USER user3 IDENTIFIED BY password3;
CREATE USER user4 IDENTIFIED BY password4;
```

Script Output | Query Result | Query Result 1

Task completed in 0,07 seconds

```
User USER1 created.

User USER2 created.

User USER3 created.

User USER4 created.
```

```
SELECT *
FROM dba_users
WHERE username IN ('USER1', 'USER2', 'USER3', 'USER4');
```

```
SELECT *
FROM dba_users
WHERE username IN ('USER1', 'USER2', 'USER3', 'USER4');
```

Script Output | Query Result | Query Result 1

SQL | All Rows Fetched: 4 in 0,008 seconds

	U...	...	PAS...	A...	LOC...	EXPIR...	DE...	T...	L...	CRE...	PRO...	INITIAL_RSRC_CON...	EX...	PASS...		AUTH...				LA...			DEFAULT_C...			
1	USER2	119	(null)	OPEN	(null)	16.06.26	USERS	TEMP	TEMP	18.1...	DEFAULT	DEFAULT_CONSUMER_GROUP	(null)	11G 12C	N	PASSWORD	N	NO	18....	N	NO	USING_NLS_COMP	NO	NO
2	USER4	124	(null)	OPEN	(null)	16.06.26	USERS	TEMP	TEMP	18.1...	DEFAULT	DEFAULT_CONSUMER_GROUP	(null)	11G 12C	N	PASSWORD	N	NO	(null)	N	NO	USING_NLS_COMP	NO	NO
3	USER3	123	(null)	OPEN	(null)	16.06.26	USERS	TEMP	TEMP	18.1...	DEFAULT	DEFAULT_CONSUMER_GROUP	(null)	11G 12C	N	PASSWORD	N	NO	(null)	N	NO	USING_NLS_COMP	NO	NO
4	USER1	122	(null)	OPEN	(null)	16.06.26	USERS	TEMP	TEMP	18.1...	DEFAULT	DEFAULT_CONSUMER_GROUP	(null)	11G 12C	N	PASSWORD	N	NO	(null)	N	NO	USING_NLS_COMP	NO	NO

11 Дать право пользователю user1 просматривать таблицу Residing и менять даты начала и окончания проживания.

```
GRANT SELECT ON Residing TO user1;

GRANT UPDATE (d_start, d_end) ON Residing TO user1;
```

```
GRANT SELECT ON Residing TO user1;

GRANT UPDATE (d_start, d_end) ON Residing TO user1;
```

Script Output | Query Result | Query Result 1

Task completed in 0,07 seconds

```
Grant succeeded.
```

12 Передать от user2 user3 право формулировать запросы к таблице Settling.

```
GRANT SELECT ON Settling TO user2 WITH GRANT OPTION;

GRANT CREATE SESSION TO user2;
ALTER USER user2 IDENTIFIED BY password2;

REVOKE SELECT ON Settling FROM user2;
```

```
GRANT SELECT ON Settling TO user2 WITH GRANT OPTION;

GRANT CREATE SESSION TO user2;
ALTER USER user2 IDENTIFIED BY password2;

REVOKE SELECT ON Settling FROM user2;
```

Script Output | Query Result | Query Result 1

Task completed in 0,07 seconds

```
Grant succeeded.
```

```
GRANT SELECT ON SYSTEM.Settling TO user3;
```

lab_3_db.sql | user_2~1 | user_4.sql | user_1.sql | SETTLING

Worksheet | Query Builder

```
GRANT SELECT ON SYSTEM.Settling TO user3;
```

Script Output

Task completed in 0,106 seconds

```
Grant succeeded.
```

13 Передать право user4 выполнять вставку и обновление таблицы Warrant, если у id_settling дата начала не равна пустоте.

```
GRANT CREATE SESSION TO user4;

CREATE OR REPLACE VIEW warrant_view AS
SELECT id_warrant, id_building, id_settling
FROM Warrant
WHERE id_settling IS NOT NULL
WITH CHECK OPTION;
```

```
CREATE OR REPLACE VIEW Warrant_View AS
SELECT id_warrant, id_building, id_settling
FROM Warrant
WHERE id_settling IS NOT NULL
WITH CHECK OPTION;
```

Script Output | Query Result

Task completed in 0,035 seconds

```
View WARRANT_VIEW created.
```

```
GRANT INSERT, UPDATE ON warrant_view TO user4;
```

```
GRANT INSERT, UPDATE ON warrant_view TO user4;
```

Script Output | Query Result | Query Result 1

Task completed in 0,07 seconds

```
Grant succeeded.
```

lab_3_db.sql | user_2~1 | user_4.sql | user_1.sql | SETTLING

Worksheet | Query Builder

```
--дата не пустое
INSERT INTO SYSTEM.Warrant_View (id_warrant, id_building, id_settling)
VALUES (409, 17, 50);

--дата пустое
INSERT INTO SYSTEM.Warrant_View (id_warrant, id_building, id_settling)
VALUES (410, 18, NULL);
```

Script Output

Task completed in 0,071 seconds

```
1 row inserted.


Error starting at line : 6 in command -
INSERT INTO SYSTEM.Warrant_View (id_warrant, id_building, id_settling)
VALUES (410, 18, NULL)
Error at Command Line : 6 Column : 20
Error report -
SQL Error: ORA-01402: представление WITH CHECK OPTION не соответствует фразе WHERE

https://docs.oracle.com/error-help/db/ora-01402/01402. 00000 -  "view WITH CHECK OPTION where-clause violation"
*Cause:    An INSERT or UPDATE statement was attempted on a view
           created with the CHECK OPTION. This would have resulted in the
           creation of a row that would not satisfy the view's WHERE
           clause.
*Action:   Examine the view's WHERE clause in the dictionary
           table VIEWS. If the current view does not have the CHECK
           OPTION, then its FROM clause must reference a second view that
           is defined using the CHECK OPTION. The second view's WHERE
           clause must also be satisfied by any INSERT or UPDATE
           statements. To insert the row, it may be necessary to insert
           it directly into the underlying table, rather than through the
           view.

More Details :
https://docs.oracle.com/error-help/db/ora-01402/
```

14 Разрешить user1 просматривать все Проживания кроме тех, которые не имеют даты

```
GRANT CREATE SESSION TO user1;

CREATE OR REPLACE VIEW Residing_With_EndDate AS
SELECT * FROM Residing WHERE d_end IS NOT NULL;

GRANT SELECT ON Residing_With_EndDate TO user1;
```

```
GRANT SELECT ON Settling TO user2 WITH GRANT OPTION;

GRANT CREATE SESSION TO user2;
ALTER USER user2 IDENTIFIED BY password2;

REVOKE SELECT ON Settling FROM user2;
```

Script Output | Query Result | Query Result 1

Task completed in 0,07 seconds

```
Grant succeeded.
```

lab_3_db.sql | user_2~1 | user_4.sql | user_1.sql | SETTLING

Worksheet | Query Builder

```
SELECT id_residing, d_start, d_end
FROM SYSTEM.Residing_With_EndDate
ORDER BY id_residing;
```

Query Result

SQL | All Rows Fetched: 4 in 0,002 seconds

	ID_RESIDING	D_START	D_END
1	301	02.08.11	30.09.11
2	302	15.08.11	15.09.11
3	305	16.09.11	30.09.11
4	307	01.08.11	01.06.12

II. Вывод

В этой лабораторной работе я успешно создал базу данных для учета проживания в общежитиях.

Что было сделано:

1. Создал 5 связанных таблиц с данными о зданиях, помещениях, ордерах на проживание и самих проживающих.

2. Настроил защиту данных - создал 4 пользователей с разными правами:
   - User1 может только смотреть данные о проживании и менять даты
   - User2 и User3 могут просматривать информацию о заселениях
   - User4 может добавлять новые ордера, но только с заполненными данными

Результат: База данных надежно защищена, разные пользователи имеют доступ только к нужной им информации, а данные сохраняются в целости и сохранности.

```
--1--
CREATE OR REPLACE PROCEDURE settle_person(
    p_id_warrant IN INTEGER,
    p_id_premises IN INTEGER,
    p_id_bas_sal_date IN INTEGER,
    p_d_start IN DATE DEFAULT SYSDATE
)
IS
    v_current_residing_id INTEGER;
    v_free_premises_id INTEGER;
    v_building_id INTEGER;
BEGIN
```

```
    -- Пробуем найти активное проживание
    BEGIN
        SELECT id_residing INTO v_current_residing_id
        FROM Residing
        WHERE id_warrant = p_id_warrant
          AND d_end IS NULL
          AND ROWNUM = 1;

        -- Выселяем
        UPDATE Residing
        SET d_end = p_d_start
        WHERE id_residing = v_current_residing_id;

        -- Заселяем в новую
        INSERT INTO Residing (id_residing, id_warrant, id_premises, d_start,
id_bas_sal_date)
        VALUES (
            (SELECT COALESCE(MAX(id_residing), 0) + 1 FROM Residing),
            p_id_warrant,
            p_id_premises,
            p_d_start,
            p_id_bas_sal_date
        );

    EXCEPTION
        WHEN NO_DATA_FOUND THEN
            -- Если проживания нет, ищем здание по ордеру
            SELECT id_building INTO v_building_id
            FROM Warrant
            WHERE id_warrant = p_id_warrant;

            -- Ищем свободную комнату
            SELECT id_premises INTO v_free_premises_id
            FROM Premises p
            WHERE p.id_building = v_building_id
              AND NOT EXISTS (
                  SELECT 1
                  FROM Residing r
                  WHERE r.id_premises = p.id_premises
                    AND (r.d_end IS NULL OR r.d_end > p_d_start)
                    AND r.d_start <= p_d_start
              )
              AND ROWNUM = 1;
```

```
        INSERT INTO Residing (id_residing, id_warrant, id_premises, d_start,
id_bas_sal_date)
        VALUES (
          (SELECT COALESCE(MAX(id_residing), 0) + 1 FROM Residing),
          p_id_warrant,
          v_free_premises_id,
          p_d_start,
          p_id_bas_sal_date
        );
  END;
END settle_person;
/
--2--
CREATE OR REPLACE FUNCTION count_free_premises(
  p_id_building IN INTEGER,
  p_date IN DATE DEFAULT SYSDATE
) RETURN INTEGER
IS
  v_total_premises INTEGER;
  v_occupied_premises INTEGER;
BEGIN
  -- Общее количество комнат в здании
  SELECT COUNT(*) INTO v_total_premises
  FROM Premises
  WHERE id_building = p_id_building;

  -- Количество занятых комнат на дату
  SELECT COUNT(DISTINCT r.id_premises) INTO v_occupied_premises
  FROM Residing r
  JOIN Premises p ON r.id_premises = p.id_premises
  WHERE p.id_building = p_id_building
    AND r.d_start <= p_date
    AND (r.d_end IS NULL OR r.d_end > p_date);

  RETURN v_total_premises - v_occupied_premises;
END count_free_premises;

--3--
CREATE OR REPLACE TRIGGER trg_residing_bas_sal_date
BEFORE INSERT ON Residing
FOR EACH ROW
BEGIN
  IF :NEW.id_bas_sal_date IS NULL THEN
     :NEW.id_bas_sal_date := TO_NUMBER(TO_CHAR(SYSDATE,
'YYYYMMDD'));
```

```
    END IF;
END;

--4--
-- Создаем последовательность
CREATE SEQUENCE seq_settling_id
START WITH 53
INCREMENT BY 1
NOCACHE
NOMAXVALUE;

-- Триггер для автоматической установки id_settling
CREATE OR REPLACE TRIGGER trg_settling_id
BEFORE INSERT ON Settling
FOR EACH ROW
BEGIN
    IF :NEW.id_settling IS NULL THEN
        SELECT seq_settling_id.NEXTVAL INTO :NEW.id_settling FROM DUAL;
    END IF;
END;

--5--
ALTER TABLE Warrant ADD Kol_war NUMBER DEFAULT 0;

CREATE OR REPLACE TRIGGER trg_increment_kol_war
AFTER INSERT ON Settling
FOR EACH ROW
BEGIN
    UPDATE Warrant
    SET Kol_war = NVL(Kol_war, 0) + 1
    WHERE id_settling = :NEW.id_settling;
END;


--TEST--
--1--
-- Посмотрим, кто сейчас живет в комнате №1 по ордеру 400
SELECT * FROM Residing WHERE id_warrant = 400 AND d_end IS NULL;

-- Выполняем тест
BEGIN
    -- Переселяем ордер 400 в комнату 1
    -- 99 - это значение для id_bas_sal_date
    settle_person(400, 1, 99, SYSDATE);
    COMMIT;
```

```
END;
/

-- Смотрим результат
SELECT id_residing, id_warrant, id_premises, d_start, d_end, id_bas_sal_date
FROM Residing
WHERE id_warrant = 400
ORDER BY d_start DESC;

--2--
SELECT
   'On Date 01.01.2035' as test_case,
   count_free_premises(17, TO_DATE('01.01.2035', 'DD.MM.YYYY')) as
free_rooms
FROM DUAL;

--3--
--текущая дата по каденд
SELECT TO_CHAR(SYSDATE, 'YYYYMMDD') AS expected_value FROM
DUAL;

SELECT * FROM Residing;

-- Вставляем запись, "забыв" про поле с датой
INSERT INTO Residing (id_residing, id_warrant, id_premises, d_start)
VALUES (
   (SELECT MAX(id_residing) + 1 FROM Residing),
   400,
   1,
   SYSDATE
);
COMMIT;

SELECT * FROM Residing;

--4--
--Смотрим текущий максимум в таблице
SELECT MAX(id_settling) AS max_id_now FROM Settling;

INSERT INTO Settling (id_e, d_start)
VALUES ('999', SYSDATE);

-- Обязательно фиксируем, иначе данные не запишутся
COMMIT;
```

```
SELECT id_settling, id_e, d_start
FROM Settling
WHERE id_e = '999';
```

--5--

----------333333

```
CREATE TABLE Residing (

id_residing INTEGER PRIMARY KEY,

id_warrant INTEGER,

id_premises INTEGER,

d_start DATE NOT NULL,

d_end DATE,

id_bas_sal_date INTEGER NOT NULL

);

CREATE TABLE Warrant (

    id_warrant INTEGER PRIMARY KEY,

    id_building INTEGER,

    id_settling INTEGER

);

CREATE TABLE Building (

    id_building INTEGER PRIMARY KEY,

    b_name CHAR(100),

    b_start DATE NOT NULL,
```

```
    b_note CHAR(100)
);
CREATE TABLE Premises (
    id_premises INTEGER PRIMARY KEY,
    p_name VARCHAR(30) NOT NULL,
    p_d_start DATE NOT NULL,
    id_building INTEGER
);
CREATE TABLE Settling (
    id_settling INTEGER PRIMARY KEY,
    id_e VARCHAR(30) NOT NULL,
    d_start DATE
);
ALTER TABLE  Residing
ADD CONSTRAINT fk_residing_warrant
FOREIGN KEY (id_warrant) REFERENCES Warrant(id_warrant);

ALTER TABLE Residing
ADD CONSTRAINT fk_residing_premises
FOREIGN KEY (id_premises) REFERENCES Premises(id_premises);

ALTER TABLE Warrant
ADD CONSTRAINT fk_warrant_building
```

```
FOREIGN KEY (id_building) REFERENCES Building(id_building);


ALTER TABLE Premises

ADD CONSTRAINT fk_premises_building

FOREIGN KEY (id_building) REFERENCES Building(id_building);


ALTER TABLE Warrant

ADD CONSTRAINT fk_warrant_settling

FOREIGN KEY (id_settling) REFERENCES Settling(id_settling);

ALTER TABLE Warrant

ADD CONSTRAINT check_warrant_gt_settling

CHECK (id_warrant > id_settling);

ALTER TABLE Building

MODIFY b_name CHAR(10) DEFAULT '_' NOT NULL;

INSERT INTO Building (id_building, b_name, b_note, b_start) VALUES

(11, 'К-1', 'Здание учебного корпуса № 1',
TO_DATE('01.01.2010','DD.MM.YYYY'));


INSERT INTO Building (id_building, b_name, b_note, b_start) VALUES

(17, 'ЗО-1', 'Здание общежития № 1', TO_DATE('01.01.2010',
'DD.MM.YYYY'));


INSERT INTO Building (id_building, b_name, b_note, b_start) VALUES
```

(18, 'ЗО-2', 'Здание общежития № 2', TO_DATE('01.01.2010', 'DD.MM.YYYY'));

INSERT INTO Settling (id_settling, id_e, d_start) VALUES

(50, '1', TO_DATE('01.08.2011', 'DD.MM.YYYY'));

INSERT INTO Settling (id_settling, id_e, d_start) VALUES

(51, '2', TO_DATE('01.09.2011', 'DD.MM.YYYY'));

INSERT INTO Settling (id_settling, id_e, d_start) VALUES

(52, '4', TO_DATE('15.08.2011', 'DD.MM.YYYY'));

ALTER TABLE Premises ADD p_area NUMBER;

INSERT INTO Premises (id_premises, p_name, p_d_start, p_area, id_building) VALUES

(1, '110', TO_DATE('22.06.2010', 'DD.MM.YYYY'), 2, 17);

INSERT INTO Premises (id_premises, p_name, p_d_start, p_area, id_building) VALUES

(2, '112', TO_DATE('22.06.2010', 'DD.MM.YYYY'), 5, 18);

INSERT INTO Premises (id_premises, p_name, p_d_start, p_area, id_building) VALUES

(4, '115', TO_DATE('22.06.2010', 'DD.MM.YYYY'), 2, 18);

INSERT INTO Premises (id_premises, p_name, p_d_start, p_area, id_building) VALUES

(5, '1110', TO_DATE('22.06.2010', 'DD.MM.YYYY'), 3, 17);

INSERT INTO Premises (id_premises, p_name, p_d_start, p_area, id_building) VALUES

(7, '211', TO_DATE('22.05.2010', 'DD.MM.YYYY'), 4, 17);

INSERT INTO Premises (id_premises, p_name, p_d_start, p_area, id_building) VALUES

(8, '1111', TO_DATE('22.05.2010', 'DD.MM.YYYY'), 50, 11);

INSERT INTO Warrant (id_warrant, id_building, id_settling) VALUES

(400, 17, 50);

INSERT INTO Warrant (id_warrant, id_building, id_settling) VALUES

(404, 17, 52);

INSERT INTO Warrant (id_warrant, id_building, id_settling) VALUES

(406, 17, 51);

INSERT INTO Warrant (id_warrant, id_building, id_settling) VALUES

(404, 18, 52);

INSERT INTO Residing (id_residing, id_warrant, id_premises, d_start, d_end, id_bas_sal_date)

VALUES (301, 400, 5, TO_DATE('02.08.2011', 'DD.MM.YYYY'), TO_DATE('30.09.2011', 'DD.MM.YYYY'), 90);

```
INSERT INTO Residing (id_residing, id_warrant, id_premises, d_start, d_end,
id_bas_sal_date)

VALUES (302, 404, 7, TO_DATE('15.08.2011', 'DD.MM.YYYY'),
TO_DATE('15.09.2011', 'DD.MM.YYYY'), 91);


INSERT INTO Residing (id_residing, id_warrant, id_premises, d_start, d_end,
id_bas_sal_date)

VALUES (305, 404, 2, TO_DATE('16.09.2011', 'DD.MM.YYYY'),
TO_DATE('30.09.2011', 'DD.MM.YYYY'), 92);


INSERT INTO Residing (id_residing, id_warrant, id_premises, d_start, d_end,
id_bas_sal_date)

VALUES (306, 400, 1, TO_DATE('01.10.2011', 'DD.MM.YYYY'), NULL, 93);


INSERT INTO Residing (id_residing, id_warrant, id_premises, d_start, d_end,
id_bas_sal_date)

VALUES (307, 406, 1, TO_DATE('01.08.2011', 'DD.MM.YYYY'), NULL, 94);

UPDATE Residing

SET d_end = TO_DATE('01.06.2012', 'DD.MM.YYYY')

WHERE id_warrant = 406;
```